

# PHYS 224 - Physics of Music

September 1, 2026

## Basic Information

|                        |                                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Online Materials       | MyCourses/Github here                                                                                                                        |
| Book (Optional)        | “The Science of Sound” by Rossing, Moore & Wheeler, but only if very motivated                                                               |
| Credits                | 3.000(0)                                                                                                                                     |
| Instructor<br>Office   | Prof. Jonathan Sievers: jonathan.sievers@mcgill.ca<br>Rutherford 333                                                                         |
| TAs                    | Han Lei (han.lei@mail.mcgill.ca)<br>Natalia Martorella (natalia.martorella@mail.mcgill.ca)<br>Noah Kakekaspan noah.kakekaspan@mail.mcgill.ca |
| <b>Tutorials (TBD)</b> | <b>Please attend any and all of these as you see fit!</b>                                                                                    |
| TBD                    | Monday 4pm - 5pm                                                                                                                             |
| TBD                    | Tuesday 11am - 12pm                                                                                                                          |
| TBD                    | Wednesday 12:30pm - 1:30pm                                                                                                                   |
| TBD                    | Thursday 9:30am - 10:30am                                                                                                                    |

## Overview

This course comprises lectures, alternating weekly homeworks and quizzes (no midterms), TA-run study groups (help with current and past homeworks / quizzes), and a 3-hour final exam. The grading scheme is flexible, allowing students to drop up to half of the quizzes, shifting weight to the final exam, as discussed below.

## AI

Feel free to use AI as a learning tool – upload homeworks, notes, solutions, ask it questions, check it’s logic, clarify things... Let us know how you use it, what works, and what does not! It is cheaper than a tutor and more available than a TA. However, **all of your submitted work must be your own**, *not* generated by an AI.

## Homework

Weekly assignments will be given to provide additional insights, applications, and practice with the formalism developed in class.

- Assignments will be graded quickly and generously as pass (100%), partial (50%), or fail (0%), based on signs of effort on every problem.
- **No points will be deducted for incorrect homework answers.** Please *try* each problem, whether they fail or succeed, and *ask* TAs / me for help! Trying, failing, and *then* figuring out the solution is ideal for learning.
- Feedback is provided through

- Posted homework solutions
- Tutorials
- **Study groups are the most efficient times to work on assignments.**
  - You are free to ask us how to do past *and present* homework problems (!).
  - You are free to attend any and all study group sections.
- Homework must be uploaded by the due date at <https://crowdmark.com/>. No late homework will be accepted.
- **Please do the homework:**
  - It's easy points, and 10% is the difference between F and C (because McGill).
  - They provide practice, automation, and immediate feedback.
  - Historically, everyone attempting all homeworks received a good final grade.
  - Historically, everyone handing in 1/3 or fewer did *very poorly*.
  - They are fun...? They are why you're here?

## Quizzes

We will have routine quizzes to provide continuous feedback and practice for the final exam. Up to half of these can be dropped, but the weight of dropped quizzes will be shifted to the final (see below), so it is in your best interest to try all quizzes, **especially if you think you'll bomb it**: if you struggle and fail a problem, then look at / discuss the solutions, chances are you'll get it next time, and this is the best way to learn. Note also it is possible to get a B without attending the final in this course!

- Quizzes will be typically given during the first 30 minutes of the lecture, with one short problem focusing on the most recent homework, and one “review” problem from previous homeworks.
- Due to the flexible grading scheme, **no make-up quizzes** will be administered.

## Final Exam

We will have a 3-hour cumulative final exam comprising several expanded quiz-style problems, covering only the concepts practiced in the homeworks and quizzes (focusing heavily on that of the quizzes).

## Course Grading Scheme

The overall grade will be determined by automatically selecting the optimal weighting from the following scheme:

- Homework: 10%, graded easily pass / fail
- Quizzes: 60% - 30%, depending on how many quizzes are dropped
- Final Exam: 30% - 60%, depending on how many quizzes are dropped

## Dates (to be updated - boilerplate only right now)

| Item         | Date    | TA   |
|--------------|---------|------|
| Assignment 1 | Sep 10  | Minh |
| Quiz 1       | Sept 17 | All  |
| Assignment 2 | Sep 24  | Kate |
| Quiz 2       | Oct 1   | All  |
| Assignment 3 | Oct 8   | Minh |
| Quiz 3       | Oct 22  | All  |
| Assignment 4 | Oct 29  | Kate |
| Quiz 4       | Nov 5   | All  |
| Assignment 5 | Nov 12  | Minh |
| Quiz 5       | Nov 19  | All  |
| Assignment 6 | Nov 26  | Kate |
| Quiz 6       | Dec 1   | All  |

## Quiz / Exam Rules and Strategies

Students may bring one 8.5" × 11" sheet of paper with whatever they like written on both sides. No electronics.

### Notes:

- Include units on all quantities.
- Electronics of any kind are strictly forbidden. If you have to “plug in numbers”, please write things like “ $\sqrt{(101 \text{ N/m}^2)/(300 \text{ kg/m}^3)}$  with units (m/s)” . Then you can make an approximate numerical answer but don’t worry about accuracy (e.g.  $\sqrt{10} = 3$  is plenty good enough).
- Put a box around your final answer to remove ambiguity.
- To save time, improve grading, and reduce algebra mistakes, you can leave your answers in terms of quantities you have previously defined / solved for (but those previous definitions must have a clear and unambiguous box around them).

### Effective strategies for partial credit:

- Show your work and explain your reasoning. Up to 60% credit can be earned for setting up the problem correctly.
- If you don’t know how to calculate a quantity, assign it a new symbol (including its functional dependencies if known) and proceed.
- If you know how to proceed but somehow can’t implement it, write it in words.
- Remain transparent about what you do not know. Writing down wrong / irrelevant information reduces partial credit.

## Course Outline (Subject to Change)

- Air and sound waves
- Frequency and pitch, musical intervals, scales, and tuning, harmonics, perception
- Superposition, fourier and timbre, perception
- Loudness and the decibel scale, perception
- Wave diffraction, directionality, and reflection in strings, fluids, and pipes

- Resonance in pipes, chambers, strings, bars, and membranes
- Time permitting:
  - Feedback
  - The human voice
  - Room reverberation and acoustics
  - Why there is no such thing as a perfectly tuned piano
  - Anything else you're interested in?

## Prerequisites

- Physics: force, momentum, Newton's Laws
- **Mathematics: algebra, logarithms, some trigonometry**
- Music: none, but knowledge of scales and intervals will be useful.

## Additional Information

1. My goal is to provide an inclusive learning environment. If you experience barriers to learning in this course, do not hesitate to discuss them with me and Student Accessibility and Achievement, 514-398-6009.
2. You are encouraged to discuss the physics and methods of homework problems with each other, the TA's and me. However, you must perform all calculations and write up solutions on your own, even if we / they give you the answer directly.
3. You can use any external software and / or websites for assignments (as long as you don't directly use AI for your answers).
4. Course structure is subject to change, and any changes will be announced during lecture.

## McGill Statements and Fine Print

1. In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French written work that is to be graded. This does not apply to courses in which acquiring proficiency in a language is one of the objectives. (Approved by Senate on 21 January 2009). Conformément à la Charte des droits de l'étudiant de l'Université McGill, chaque étudiant a le droit de soumettre en français ou en anglais tout travail écrit devant être noté, sauf dans le cas des cours dont l'un des objets est la maîtrise d'une langue. (Énoncé approuvé par le Sénat le 21 janvier 2009)
2. McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Code of Student Conduct and Disciplinary Procedures. (Approved by Senate on 29 January 2003) (See McGill's guide to academic honesty for more information). L'université McGill attache une haute importance à l'honnêteté académique. Il incombe par conséquent à tous les étudiants de comprendre ce que l'on entend par tricherie, plagiat et autres infractions académiques, ainsi que les conséquences que peuvent avoir de telles actions, selon le Code de conduite de l'étudiant et procédures disciplinaires. (Énoncé approuvé par le Sénat le 29 janvier 2003) (pour de plus amples renseignements, veuillez consulter le guide pour l'honnêteté académique de McGill.)
3. In the event of extraordinary circumstances beyond the University's control, the content and/or evaluation scheme in this course is subject to change.
4. Note that to support academic integrity, your assignments may be submitted to text-matching or other appropriate software (e.g., formula-, equation-, and graph-matching).

5. © Instructor-generated course materials (e.g., handouts, notes, summaries, exam questions, etc.) are protected by law and may not be copied or distributed in any form or in any medium without explicit permission of the instructor. Note that infringements of copyright can be subject to follow up by the University under the Code of Student Conduct and Disciplinary Procedures.
6. McGill University is on land which has long served as a site of meeting and exchange amongst Indigenous peoples, including the Haudenosaunee and Anishinabeg nations. We acknowledge and thank the diverse Indigenous people whose footsteps have marked this territory on which peoples of the world now gather.
7. Individual grades on any assignment, report, quiz, or exam may only be disputed for up to 30 days after these grades are disseminated.